



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2025-26/opn/380255

Date:10/08/2025(19:43:31)

Subject:Controlled Fast Cooling Down of KWU Steam Turbines

LMI on "Controlled Fast Cooling Down of KWU Steam Turbines" LMI/OGN/OPS/MECH/042

The following LMI has been revised in line with revised OGN vide "COS-ISO-00-OGN/OPS/MECH/042, Rev. No.: 3, May 2025

Revision details:

- Under clause 5.2.1 points (e)
- Under clause 5.2.2 points (i)
- Under clause 5.2.3 points (e)
- Annexure 1
- Annexure 5

2.1 RECORDS OF REVISION:

		Revision 3
Cl no.	Revision 2	
All Clause		Responsibilities of carrying out different activities included
Clause	For spring mounted TG decks/condenser, both	For spring mounted TG

5.2.1(e)	CW pumps are to be kept in service initially. One CW pump can be stopped once HP casing / rotor temperature comes below 325 Deg C and 2nd CW pump may be stopped after stoppage of barring gear. For other make/model steam turbines, recommendations of OEM is to be followed.	decks/condenser of KWU design sets, both CW pumps are to be kept in service initially. One CW pump can be stopped once HP casing / rotor temperature comes below 325 Deg C and 2nd CW pump may be stopped after stoppage of barring gear. For other make/model steam turbines, recommendation of OEM is to be followed. Any deviation to the above procedure shall be mutually decided between site & COS.
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Clause 5.2.2(i)	Ensure HP casing drain and all MAL drains in opened condition in local and their manual isolating valves are opened. (SLC drains to be kept in manual)	Ensure closing of HP casing drain (MAL-22) and drain between stop & control valve (MAL-11 & 12) and all remaining MAL drains in opened condition in local and their manual isolating valves are opened. (SLC drains to be kept in manual). The Hot Air from HP Casing will be evacuated to condenser through HPT Evacuation valve and /or CRH line drain to flash tank before NRV.
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Clause 5.2.3(e)	Any deviation to the above procedure shall be mutually decided between site & COS.
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Annexure-1	Scheme for fast cooling down of Steam Turbine updated
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Annexure-5	Checklist for Controlled Fast Cooling of KWU Steam Turbines added.
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Put up for kind approval please

Submitted by:

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On:10/08/2025(19:43:31) **Ref:**SMTPP/OPERATION/2025-26/opn/380255

Note No:2

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On:13/08/2025(04:46:03) **Ref:**SMTPP/OPERATION/2025-26/opn/380255

Note No:3

Sir

As discussed ,change has been done

Put up for the kind approval please

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On:13/08/2025(21:05:34) **Ref:**SMTPP/OPERATION/2025-26/opn/380255

Note No:4

LMI revised as per OGN-COS-ISO-00-OGN/OPS/MECH/042,Rev. No.3, May 2025

May please approve

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On:13/08/2025(23:24:35) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:5

Please review.

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Note No:6

Sir,

The change has been made as per the your suggestion.

Put up for kind approval please

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On:23/08/2025(14:44:25) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:7

Checked and forwarded.

Submitted by:

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On:23/08/2025(19:03:25) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:8

Please review.

Submitted by:

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On:29/08/2025(19:10:25) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:9

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

Submitted by:

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On:30/08/2025(11:22:20) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:10

Please make necessary format changes as per standard format.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

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On:02/09/2025(19:18:39) **Ref:**SMTPP/OPERATION/2025-26/opn/380255

Note No:11

The modification has been carried out in accordance with the standard format of Simhadri LMI, as per the guidelines outlined in **LMI/OGN/GEN/002 Rev:04** (Guidelines for Issue of LMIs as a Controlled Document), available on the Simhadri EEMG LAN.

Kindly review and confirm if any further action is required.

Please find the attachment "**LMI guidelines for issue of LMI as controlled doc**" accordingly for your kind reference.

Submitted by:

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On:17/09/2025(00:06:15) **Ref:**SMTPP/OPERATION/2025-26/opn/380255

Note No:12

Submitted for further review and approval.

Submitted by:

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On:17/09/2025(14:27:43) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:13

LMI is in line with the revised OGN.

Forwarded for further review.

Submitted by:

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On:20/09/2025(17:56:59) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:14

please review

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On:20/09/2025(18:21:21) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:15

LMI reviewed and is in line with the revised OGN.

Submitted by:

के पी रमेश - उप महाप्रबंधक

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On:20/09/2025(18:54:48) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:16

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On:22/09/2025(09:43:05) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:17

LMI is in line with the revised OGN. May please be approved.

Submitted by:

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On:22/09/2025(10:02:35) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:18

May please approve.

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Note No:19

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On:24/09/2025(11:12:58) **Ref:**SMTTP/OPERATION/2025-26/opn/380255

Note No:20

Approved & Submitted by:

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On:24/09/2025(11:38:34) **Ref:**SMTPP/OPERATION/2025-26/opn/380255

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Controlled Fast Cooling Down of KWU Steam Turbines

LMI/OGN/OPS/MECH/042 Rev: 03

JULY - 2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Controlled Fast Cooling Down of KWU Steam Turbines
2	LMI No.	LMI/OGN/OPS/MECH/042 Rev:03
3	Reference Document	COS-ISO-00-OGN/OPS/MECH/042/ Rev:03 May:2025
4	Superseded Document	LMI/SMSTPS/OIN/GEN/002 Rev:03 April 22
5	Prepared By	Name: KM POOJA, Engineer (O)
6	Checked By	Name: Somnath Bhattacharya, AGM(O)
7	Checked By	Name: Navin Kishore, AGM(EEMG)
8	Checked By	Name: Nilanjan Mandal, AGM(TMD)
9	Checked By	Name: Hemendra Kumar Verma, GM(Maint)
10	Reviewed By	Name: Ahmad Kaunain Raza GM (O&M)
11	Approved By	Name: Sameer Sharma ED HOP (Simhadri)

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
LMI No.	LMI/OIN/GEN/002 Rev:04

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TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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1.0 Introduction

This document aims to provide guidance and procedure for controlled fast cooling of KWU design turbines during shut down for turbine overhaul without stressing turbine internals and components. This will result in reduction of the cooling period and thereby quicker starting of maintenance work after the stoppage of barring gear. In KWU machine, the average time to stop barring gear usually takes 5 to 6 days in old generation 500 MW units & 8 to 9 days in new generation 500 MW units after taking out machine from bar. The overhaul works starts after stopping barring gear. By controlled fast cooling of steam turbine, barring gear can be stopped on an average within 74 hours after taking out machine of bar to start maintenance/overhaul work. In addition to that sometimes TG barring speed reduced to zero or barring gear stalled due to some inadvertent issues after unit trippings. The controlled fast cooling of turbine rotors have been exercised to ensure earliest revival of unit..

2.0 Superseded Documents

Rev 04 Oct'2022 – LMI has been revised as per COS-ISO-00-OGN/OPS/MECH/042, Rev.No.: 2, Sep. 2022

- Under clause 5.1 (c)
- Under clause 5.2.1 points (e), (f), (h), (I), (J)
- Under clause 5.2.2 points (m), (o), (p)

Rev 12 July'2025 – LMI has been revised as per COS-ISO-00-OGN/OPS/MECH/042, Rev.No.: 3, May 2025

- Under clause 5.2.1 points (f)
- Under clause 5.2.2 points (i)
- Under clause 5.2.3 points (e)
- Annexure 1
- Annexure 5

3.0 Details of Revision

Cl no.	Revision 2	Revision 3
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All Clause		Responsibilities of carrying out different activities included
Clause 5.2.1(f)	For spring mounted TG decks/condenser, both CW pumps are to be kept in service initially. One CW pump can be stopped once HP casing / rotor temperature comes below 325 Deg C and 2nd CW pump may be stopped after stoppage of barring gear. For other make/model steam turbines, recommendations of OEM is to be followed.	For spring mounted TG decks/condenser of KWU design sets, both CW pumps are to be kept in service initially. One CW pump can be stopped once HP casing / rotor temperature comes below 325 Deg C and 2nd CW pump may be stopped after stoppage of barring gear. For other make/model steam turbines, recommendation of OEM is to be followed. Any deviation to the above procedure shall be mutually decided between site & COS.
Clause 5.2.2(i)	Ensure HP casing drain and all MAL drains in opened condition in local and their manual isolating valves are opened. (SLC drains to be kept in manual)	Ensure closing of HP casing drain (MAL-22) and drain between stop & control valve (MAL-11 & 12) and all remaining MAL drains in opened condition in local and their manual isolating valves are opened. (SLC drains to be kept in manual). The Hot Air from HP Casing will be evacuated to condenser through HPT Evacuation valve and /or CRH line drain to flash tank before NRV.
Clause 5.2.3(e)		Any deviation to the above procedure shall be mutually decided between site & COS.

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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Annexure-1		Scheme for fast cooling down of Steam Turbine updated
Annexure-5		Checklist for Controlled Fast Cooling of KWU Steam Turbines added.

3.0 SCOPE

This is applicable to 500 MW steam turbines installed at NTPC Simhadri for stage-1&2 .

4.0 CONTROLLED FAST COOLING DOWN PROCEDURE

It is recommended that Controlled fast cooling of turbine may be done in two stages. In first stage unit parameters are to be lowered by reducing load and steam parameters. In the second stage after de-synchronization of the machine, turbine is to be cooled by natural cooling till HP casing temperature drops to 300°C. Thereafter turbine is to be cooled by entering air through HPCV and temperature drop rate is to be maintained at 5- 6 °C per hour by vacuum & HP control valve modulation or HP Evacuation valve.

5.0 UNIT HISTORY & OPERATIONAL PROBLEMS

Before deciding on forced cooling following should be analyzed

- History of the unit, trouble shootings, problems, operational parameters, & turning gear problems if any should be logged.
- If any deviation in parameters such as high diff. top/bottom temp., differential expansions, axial shift, bearing temperature, bearing & shaft vibrations is observed during operation, reasons of such deviations are to be established and must be rectified.
- If the above deviations are beyond rectification, then this process should not be carried out.

5.1 STAGE-I : PARAMETER REDUCTION ON LOAD

- Ensure all auto loops are in service and no protection is bypassed.
- Ensure all SLCs & SGCs of AOP, JOP, turning gear, drain, and temp. Controller etc. are in auto.

Resp-UCE

Start parameter reduction six hours before planned shut down time as given below.

- Reduce load by 5% and then reduce main steam temperature as per TSE lower margins/ turbine cooling curves in steps, till HP casing temperature comes to 480 °C (or at the

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nearest value to it to avoid turbine trip on MS temp low) and load to 70%. Relative expansions & differential temperature HP/IP should not reach alarm limits.

Resp-UCE

- d) The rate of reduction of main steam temperature should not exceed 1.5 to 3°C per minute.

Resp-UCE

- e) Reduce the load up to the lowest possible allowed as per the unit operating parameters.

Resp-UCE

- f) Do not reduce load & temperature at the same time because of the danger of exceeding the permissible relative expansion and the lower margin of the TSE/TSC as well as possibility of developing transient vibrations.

Resp-UCE

Always take care that the HP exhaust temperature is not reduced to saturation temperature corresponding to the HP exhaust pressure.

- g) After reducing the HP casing temperature to 480 C, operate turbine with reduced load and steam temperatures until the lower margin of TSE/TSC does not change significantly & then desynchronize the machine.

Resp-Turb Desk Engineer

- h) Ensure Opening all MAL drains of HP cylinder, stop/control valves and extraction valves one by one & put them in auto, in order to drain out the flushing condensate which may be formed during parameter reduction.

Resp-Turb Desk Engineer

- i) Ensure start of AOP, JOP & opening of gate valve gearing as per logic.

Resp-Turb Desk Engineer

- j) Ensure that the turbine comes on turning gear and record coasting down time.

Resp-Turb Desk Engineer

5.2 STAGE-II : CONTROLLED FAST COOLING OF TURBINE

5.2.1 NATURAL COOLING OF TURBINE

- a) After depressurization of Main Steam line, ensure that Boiler Stop Valves (MS-1, MS-2) & their bypass valves are closed and isolated.

Resp-UCE

- b) Ensure all MAL drains, extraction NRV drains and SLC drains are in

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auto and opened as per logic.

Resp-Turb Desk Engineer

- c) Ensure all LP & HP heaters, TDBFP and D/A extraction valves are closed and alternate drip valves are open so that there is no drip level.
Resp-Turb Desk Engineer
- d) Ensure that HP bypass spray valves are closed & isolated. Ensure closing of HPBP Spray Block Valve & Manual isolating valve. Resp-Turb Desk Engineer
- e) Keep condenser charged from CW side and run one CEP and monitor LP Exhaust hood temperature.
Resp-UCE
- f) For spring mounted TG decks/condenser of KWU design sets, both CW pumps are to be kept in service initially. One CW pump can be stopped once HP casing / rotor temperature comes below 325 Deg C and 2nd CW pump may be stopped after stoppage of barring gear. For other make/model steam turbines, recommendations of OEM is to be followed. Any deviation to the above procedure shall be mutually decided between site & COS.
Resp-UCE
- g) Ensure running of minimum one CEP during the entire process of controlled fast cooling of turbine. Running CEP can be stopped only after stopping TG barring gear.
Resp-UCE
- h) Kill condenser vacuum and isolate turbine gland seal steam supply to turbine & stop gland steam exhaust fan within 2 hours of unit outage as per COS-ST IOM (Ref No. OS/ST/MT/COND VAC Dated 29.12.2013). Observe rate of Fall in HPT Casing / Shaft Temperature (acceptable rate of temperature drop is 5°C - 6°C per hour) and allow natural cooling till HP shaft & HP casing temperature comes down to 300°C
Resp-UCE.
- i) Ensure opening of atmospheric drains of MS/HRH/CRH header and also ensure closing of the above drains connected to flask tank/condenser. Please ensure no wet steam entry into condenser.
Resp-UCE
- j) In case there is significant change in Rate of Temperature drop

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(Deviation from the acceptable rate of 5°C - 6°C per hour), keep condenser vacuum pulled and seal steam supply in charged condition & gland steam exhaustor in service to maintain desired rate of cooling of HP shaft till HP shaft & HP casing temperature comes down to 30°C by Natural Cooling. In case there is no improvement in temperature drop deviations, then vacuum may be killed & seal steam may be isolated.

Resp-UCE

- k) If seal steam parameters could not be maintained then Kill condenser vacuum, stop turbine gland seal steam supply to turbine and stop gland steam exhaustor fan.

Resp-UCE

- l) Both CW pumps are to be kept in service initially. One CW pump can be stopped once HP casing / rotor temperature comes below 325 Deg C and 2nd CW pump may be stopped after stoppage of barring gear.

Resp-UCE

- m) Fill the hourly log sheet attached in ANNEXURE-1.

Resp-Turb Desk Engineer

5.2.2 CONTROLLED FAST COOLING MAINTAINING COOLING RATE OF 5 deg-6 deg PER HOUR BY ADMITTING AIR.

- a) Once HP casing temperature has reached 300 °C by natural cooling we can proceed to go for controlled fast cooling.

Resp-UCE/Turb Desk Engineer

- b) Ensure that the turbine is on turning gear and all Turbine stop and control valves are closed.

Resp-UCE/Turb Desk Engineer

- c) Ensure that all MAL drain lines are properly drained.

Resp-UCE/Turb Desk Engineer

- d) Open air inlet sockets between main steam stop valve and control valve of HP turbine by removing Cap nuts and Cover Flange.

Resp-TMD

- e) Cover the opening with fine wire mesh to avoid any foreign material entering inside.

Resp-TMD

Note: It is to be ensured that the air entering through the socket

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must be dry

Adequate precaution is to be taken so that water ingress is prevented through this opening.

- f) Set test valves of all stop valves in closed condition. Resp-UCE

- g) Open HP control valves and keep IP control valves in closed position with the help of test motor or by isolating IP secondary oil. Resp-UCE

- h) Ensure cleanliness of turbine gland area and pull vacuum without charging gland steam to turbine. (keep vacuum breaker closed) also throttle condenser vacuum pumps suction manual valve seeing its current Resp-UCE

Note: no insulation removal work should be carried out during fast cooling.

- i) Ensure closing of HP casing drain (MAL-22) and drain between stop & control valve (MAL11 & 12) and all remaining MAL drains in opened condition in local and their manual isolating valves are opened. (SLC drains to be kept in manual). The Hot Air from HP Casing will be evacuated to condenser through HPT Evacuation valve and /or CRH line drain to flash tank before NRV.

Resp-Turb Desk Engineer

- j) Maintain condenser vacuum -0.2 kg/cm^2 initially and observe turbovisory readings and TSE/TSC parameters.

Resp-Turb Desk Engineer

- k) Maintain condenser Vacuum and control air entry to turbine in such a way so that all the turbovisory readings and relative expansion of HP, IP & LP turbines and lower margin of TSE/TSC remain within limits.

Resp-UCE/Turb Desk Engineer

- l) Maintain temperature drop rate of 5 to 6 °C per hour by regulating the condenser vacuum and HP control valves.

Resp-UCE/Turb Desk Engineer

- m) If the cooling rate is observed to be less than 5°C per hour, increase

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condenser vacuum gradually (maximum up to -0.4 kg/cm^2) and maintain the drop rate at 5°C/hr by modulating HP control valves. If required HPT Evacuation valve to be modulated in order to maintain temperature drop rate of 5 to 6°C per hour.

Resp-UC/ Turb Desk
Engineer

- n) In case any of the parameters such as:
- Relative expansion of HP/IP/LP turbines
 - Lower margin of TSE
 - HP casing top/bottom differential temperature

are observed near the limiting values, reduce the condenser vacuum (kill if required). Normally this situation does not arise if vacuum is maintained between -0.2 to -0.4 kg/cm^2 and drop rate is maintained around 5°C to 6°C/hr .

Resp-UC

- o) Turbine insulation can be removed at casing temperature of 150°C only after stopping controlled fast cooling.

Resp- TMD

- p) After the completion of the turbine insulation removal work and the clearance from TMD, continue the fast-cooling process till the casing and shaft temperature comes down to 110°C . It has been observed that after 74 hrs of controlled fast cooling from de- synchronization with a drop rate of 5°C/hr , the turbine metal temperature, HP casing and HP shaft temperature comes down to 110°C .

Resp-UC

- q) At this stage kill condenser vacuum and stop turning gear.

Resp UC

- r) After completing controlled fast cooling process, plot curves of cooling time vs. turbine metal temperature (HP casing & HP shaft), from the collected data in the log sheets

5.2.3 Controlled Fast Cooling During TG Barring Gear Stalling And During Any-Exigency

During TG barring gear stalling and any exigency for carrying out any work in TG set (except planned OH) - Controlled Fast Cooling maintaining Cooling Rate of $5^\circ\sim 6^\circ\text{C}$ per Hour by Admitting Air

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- a) Once HP casing temperatures have reached 400 °C by natural cooling, we can proceed to go for Controlled Fast Cooling.
- b) Avoid start/stop of CW system and draining of condenser water box during TG barring gear stalling.
- c) The other steps as mentioned in SI no. 5.2.2 shall be followed.
- d) Once the machine came on barring gear, then machine shall be kept on barring gear for min 4 hours before going for turbine rolling
- e) Any deviation to the above procedure shall be mutually decided between site & COS.

Resp- HOD(O) / HOD(TMD)

6.0 Responsibility

HOD (O) is responsible for implementation of this LMI.

7.0 Reference Document

OPERATION GUIDANCE NOTE: COS-ISO-00-OGN/OPS/MECH/042, REV. No.: 2, Sep.2022

8.0 Review:

HOD (Operation) shall arrange for review of this LMI after 2 years or as necessary

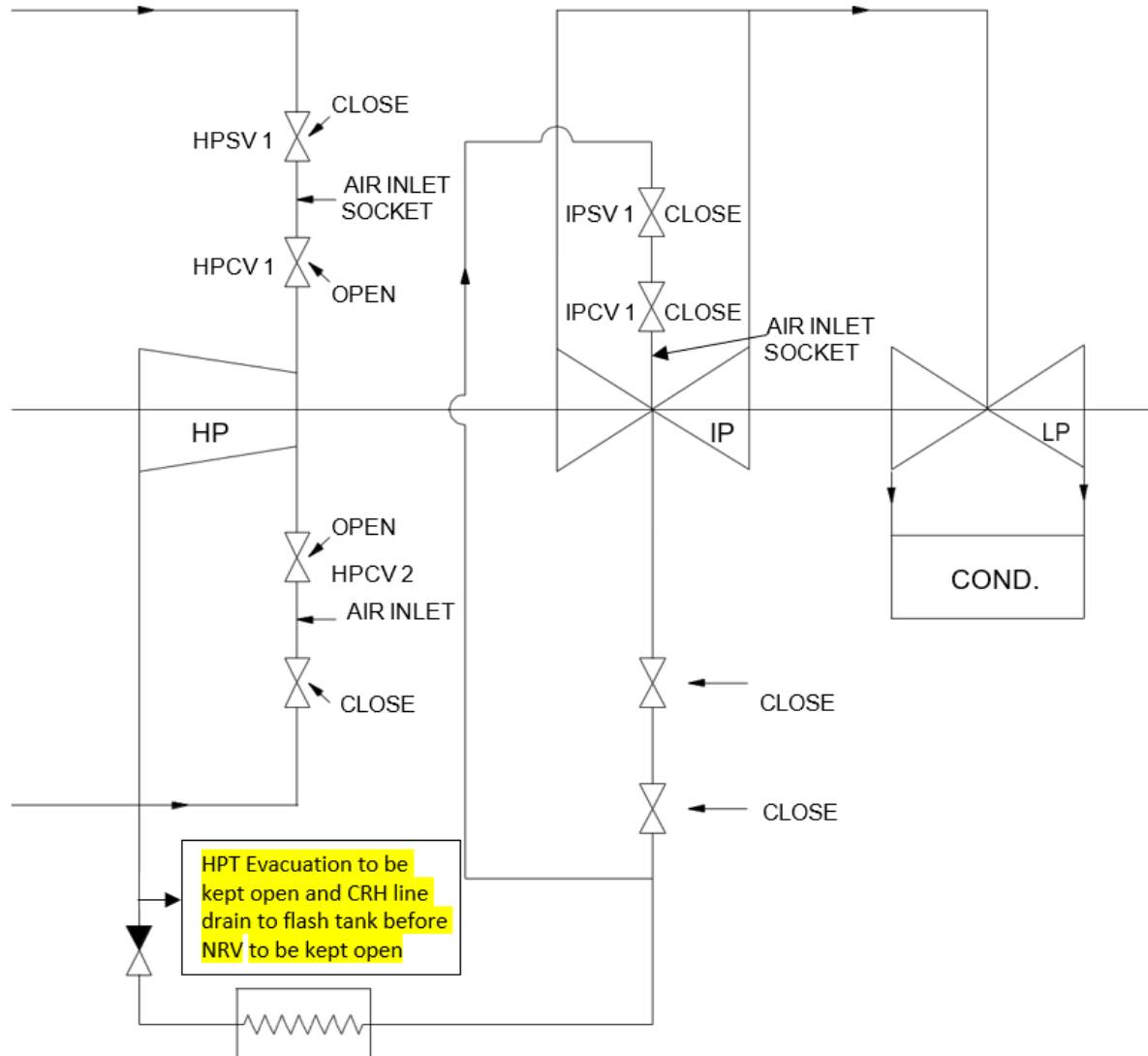
9.0 Distribution List:

Operation Dept Intranet

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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ANNEXURE-1**SCHEME FOR FAST COOLING DOWN OF STEAM TURBINE
(KWU DESIGN)**

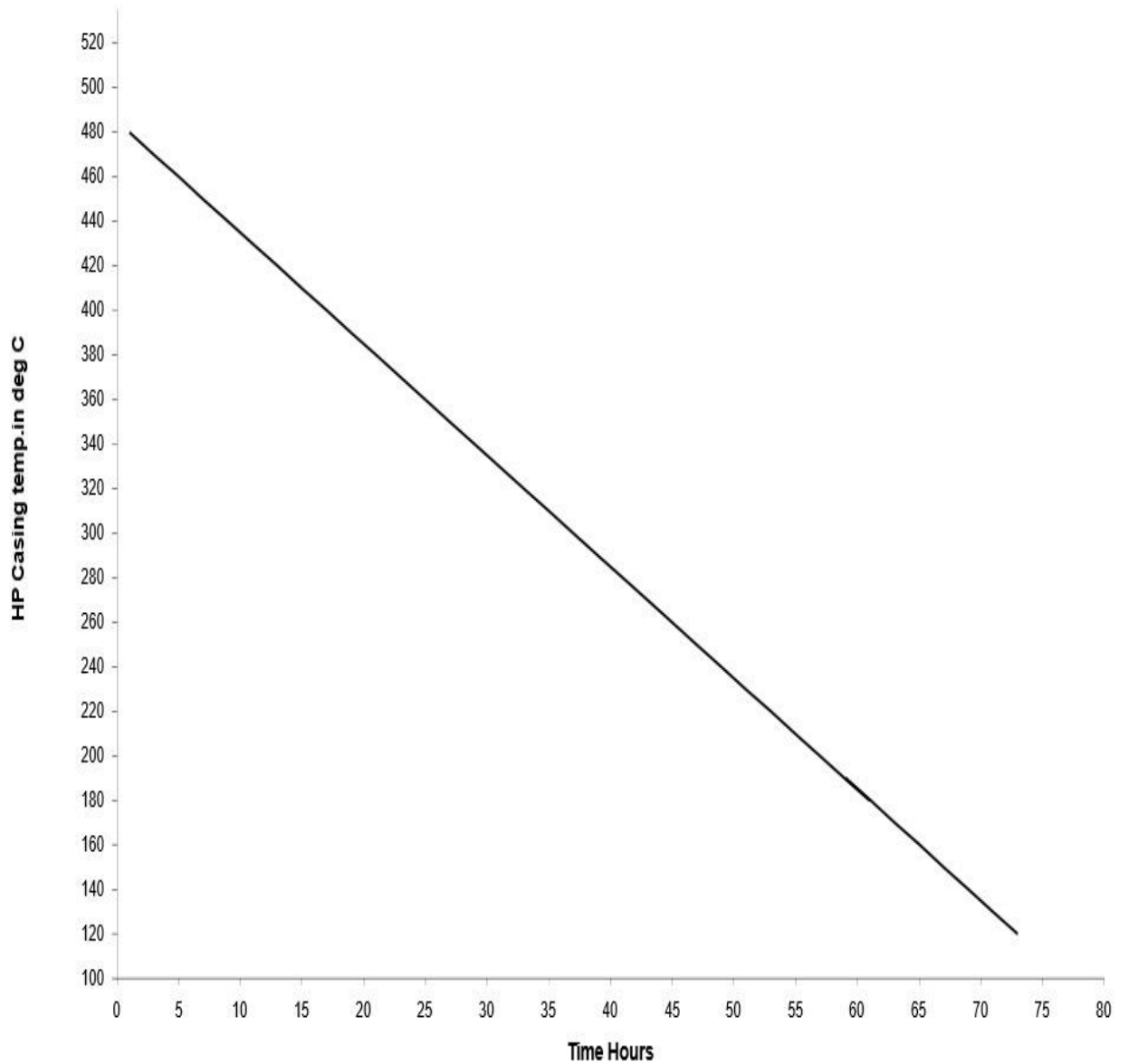
TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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ANNEXURE-2

controlled Fast Cooling Curve with temperature drop rate of 5 deg. per hour

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**ANNEXURE-3A****CONTROLLED FAST COOLING OF TURBINE LOG SHEET – 1**

STATION / PROJECT	SIZE / CAPACITY
UNIT NO.	

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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DATE -	TIME
FAST COOLING STARTS	BARRING GEAR STOPPED

Time	Hours	Load	MS Pr/Temp	CRH Pr/Temp	HRH Pr/Temp	HRH EXH Temp	Seal Steam Pr/Temp	HPCV %	IPCV %	Seal steam V/V %	LP EXH Temp	HPBP %	HPBP Down Temp	HPT Top/Bot Temp	IPT Top/ Bot Temp
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	21														
	22														
Time	Hours	Load	MS Pr/Temp	CRH Pr/Temp	HRH Pr/Temp	HRH EXH Temp	Seal Steam Pr/Temp	HPCV %	IPCV %	Seal steam V/V %	LP EXH Temp	HPBP %	HPBP Down Temp	HPT Top/Bot Temp	IPT Top/ Bot Temp

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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Time	Hours	Load	MS Pr/Temp	CRH Pr/Temp	HRH Pr/Temp	HRH EXH Temp	Seal Steam Pr/Temp	HPCV %	IPCV %	Seal steam V/V	LP EXH Temp	HPBP %	HPBP Down Temp	HPT Top/Bot Temp	IPT Top /Bot

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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	57														
Time	Hours	Load	MS Pr/Temp	CRH Pr/Temp	HRH Pr/Temp	HRH EXH Temp	Seal Steam Pr/Temp	HPCV %	IPCV %	Seal steam V/V %	LP EXH Temp	HPBP %	HPBP Down Temp	HPT Top/Bot Temp	IPT Top /Bot Temp
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TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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ANNEXURE-3B

CONTROLLED FAST COOLING OF TURBINE LOG SHEET – 2

STATION / PROJECT	SIZE / CAPACITY
UNIT NO.	
DATE -	TIME
FAST COOLING STARTS	BARRING GEAR STOPPED

Time	Hours	Speed	Vacuum	Overall expansion (HP) mm	Overall expansion (IP) mm	Relative expansion (IP) mm	Relative expansion (IP) mm	Relative expansion (LP) mm	Axial Shift	HP Casing inn/out Temp.	HPSHT inn/out temp	IP SHT inn/out Temp	HPSV inn/out Temp.	HPCV inn/out Temp.
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TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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	22													
Time	Hours	Speed	Vacuum	Overall expansion (HP) mm	Overall expansion (IP) mm	Relative expansion (IP) mm	Relative expansion (IP) mm	Relative expansion (LP) mm	Axial Shift	HP Casing inn/out Temp.	H P S H T inn/out temp	IP S H T inn/out Temp	HPS V inn/out Temp.	HPC V inn/out Temp.
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Time	Hours	Speed	Vacuum	Overall expansion (HP) mm	Overall expansion (IP) mm	Relative expansion (IP) mm	Relative expansion (IP) mm	Relative expansion (LP) mm	Axial Shift	HP Casing inn/out Temp.	HPSHT inn/out temp	IP SHT inn/out Temp	HPSV inn/out Temp.	HPCV inn/out Temp.
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TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
LMI No.	LMI/OIN/GEN/002 Rev:04

ANNEXURE-4

COS-ST IOM ON KILLING OF CONDENSER VACUUM AFTER UNIT OUTAGE

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
LMI No.	LMI/OIN/GEN/002 Rev:04



CORPORATE OPERATION SERVICES, NOIDA
INTER OFFICE MEMO

FROM: GM (OS-ST) / 3117

To: GM (O&M), Vindhyachal, Korba,
Simhadri, Ramagundam, Sipat,
Talcher K, Kahalgaon, Farakka,
Dadri Th, Rihand, & Mauda

REF. NO.: OS/ST/MT/Cond vac/

DATE: 29.12.2013

Sub: Condenser Vacuum Killing after Unit trip /shut down

Barring gear stalling in the new fleet of 500 MW spring deck supported TG sets have been observed. It is experienced that after unit trip /shut down if vacuum in condenser is maintained for longer time, chances of barring gear speed falling and stalling of barring gear is increased due to various reasons i.e. gland steam temp inadequacy, condensation in seal steam line because of supply of aux steam from other units etc. In view of this it is advised to kill condenser vacuum & isolate seal steam supply to turbine if unit shut down is more than two hrs. All other check list to be adhered as per unit shut down procedure.

B Kanungo

Copy to: GM /C Vindhyachal, Korba, Simhadri, Ramagundam, Sipat, Talcher K, Kahalgaon, Farakka, Dadri Th, Rihand, & Mauda

Copy to : for kind information pf

: ED (OS)

ANNEXURE-5

CHECK LIST FOR CONTROLLED FAST COOLING OF KWU STEAM TURBINE
Ref COS-ISO-00-OGN/OPS/MECH/042, Rev. No. : 3

TITLE	Controlled Fast Cooling Down of KWU Steam Turbines
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S. No.	Check list point for consideration	Check	Remarks
1	Review Unit History - deviation in parameter, viz. high DE, Top/Bottom diff. temp., axial shift, bearing temp, brg/shaft vibration, Turning gear problem, etc.		Historically known issues to be logged and rectified
2	Known deviations to be rectified (required for Go ahead to carry out Controlled Fast Cooling)		Restriction
	Stage-I: Parameter reduction on Load (Check list)		
3	Check all auto loops, SLCs & SGCs of AOP, JOP, EOP, Turning Gear, Turbine Drain and Lube oil temperature Controller are in Auto		
4	Load reduction to start 6 hours before planned shut-down and rate to be kept as 5% keeping MS Temp as per TSE lower margins/turbine cooling curves till HP Casing Temperature comes to 480 °C and Load to 70 %		
5	Rate of reduction of MS Temperature between 1.5 °C to 3 °C per Minute		Do not reduce Load & Temperature at the same time to keep relative expansion within permissible limits
6	Operate Turbine with reduced load and steam temperatures keeping HP Casing temperature to 480 °C		Monitor TSE/TSC before desynchronizing machine
7	Check opening of all MAL drains of HP Cylinder, Stop/Control valves and extraction valves as per logic		

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8	Check Auto Start of AOP, JOP & Opening of Gate Valve gearing as per Logic		Monitor Coasting down time
9	Check Turbine comes on Barring		Monitor Barring speed is stable
	Stage-I: Controlled Fast Cooling of Turbine (Check list)		
	Part-A - Natural Cooling		
10	Check that Boiler Stop Valves and their IBP are closed and isolated after depressurization of Main Steam Line		
11	Check that all LP & HP heaters, TDBFP and D/A Extraction Valves are closed and Alternate drip valves are open		
12	Check that all MAL drains and Extraction drains are in AUTO and open as per logic		
13	Check that HPBP Spray valves are closed and isolated and Manual Blocking valve is closed		
14	Check both CW Pumps are kept in Service, for Spring mounted TG decks		
15	Check running of minimum one CEP		
16	After Unit shut down, kill Condenser Vacuum, stop Seal Steam supply to Turbine glands and stop Gland Steam Exhauster fan within 2 hours.		

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17	Check for Rate of Fall in HPT Casing temperature which should be about 5 deg to 6 deg per hour		In case of deviation, Vacuum pulling to be done with Seal Steam charged and Gland Steam exhaustor in service to maintain rate of fall in temperature
18	Continue Natural cooling of Turbine till HP Casing Temperature comes down to 300 °C(in case of Exigency Natural Cooling may be done till 400°C)		Fill hourly log sheet to record Rate of fall in temperature
	Part-B - Controlled Cooling		
19	Check Turbine is on Barring		
20	Check all MAL drain lines are properly drained by measuring drain pipe temperature using Temperature gun		
21	Open Air inlet sockets that exist in valve body of Combined HP Stop/Control Valve and IP inlet pipe and ensure that after opening is completely covered with fine wire mesh to prevent ingress of any foreign material		
22	Check that all test valves of Turbine Stop valves are in closed condition		
23	Check Oil lines of IP Secondary Oil to IP Control Valves are kept isolated with IPCV in closed position		
24	Ensure closing of HP casing drain (MAL-22) and drain between stop & control valve (MAL-11 & 12) and all remaining MAL drains are in Open condition along with their Manual isolating valves.		SLC drains to be kept in Manual
25	Ensure opening of HPT Evacuation valve and /or CRH line drain to flash tank before NRV to evacuate the Hot Air from HP Casing to condenser.		
26	Check Cleanliness of Turbine Gland area		

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27	Pull Vacuum without charging gland steam to Turbine maintaining Condenser Vacuum -0.2 kg/cm ²		Keep Turbovisory readings and TSE/TSC parameters under observation
28	Check temperature drop rate 5 °C - 6 °C per hour and maintain it by regulating the condenser vacuum and modulating HP Control Valves		Condenser Vacuum maximum up to -0.4 kg/cm ²
29	Check parameters such as - Relative Expansion of HP/IP/LP Turbines, Lower margin of TSE, HP Casing Top/Bottom Differential temperatures are within limits		In case of major anomaly, reduce/kill condenser vacuum
30	Check that temperature comes down to 110 °C by keeping Vacuum pulled condition and modulating HP Control Valves after which kill Condenser vacuum and Stop Turning Gear		